

Modeling Optimal Stock Prices in S&P 1500 Insurance Industry

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I. Abstract

Using various predictive analytics tools in R, the aim of the study is to assess the predictive power of earnings per share (EPS), book value per share (BVPS), current ratio (CR), and debt to total assets (DTA) in relation to stock price. The data of interest are companies in the insurance industry from the S&P 1500, which contains the 1,500 largest companies with 11 unique sectors and 77 industries. The insurance industry accounts for 4.15% of total S&P stock prices, which is \$63,681.19 out of \$1,534,079.

II. Introduction

The data is from the S&P 1500, which contains the 1500 largest companies with 11 unique sectors and 77 industries. This study will analyze the predictive power of EPS, BVPS, CR, and DTA to identify the optimal stock price for companies in the insurance industry. Predictive analytics tools will include both multiple linear, parametric regression and multiple nonlinear, nonparametric regression. Investors, investment bankers, and portfolio managers would be interested in the results of the study.

III. Previous Research

There is no previous research focusing on this topic.

IV. Methodology

Eqn 1 (Functional specification) Price $\sim$ EPS (+) , BVPS (+), CR (+), DTA (-)

Eqn 2 (Population Regression) Price_i = α + β_1 EPS + β_2 BVPS + β_3 CR + β_4 DTA + ϵ_i

$$\text{Eqn 3 (Sample regression)} \text{ Price}_i = a + b_1\text{EPS} + b_2\text{BVPS} + b_3\text{CR} + b_4\text{DTA}$$

V. Results

Fig. 1 Hist of Price

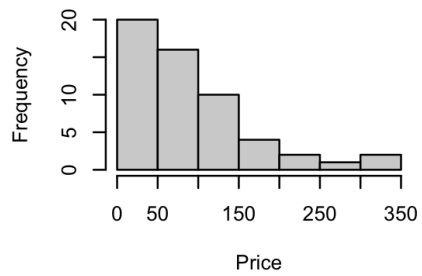


Fig. 2 Hist of EPS

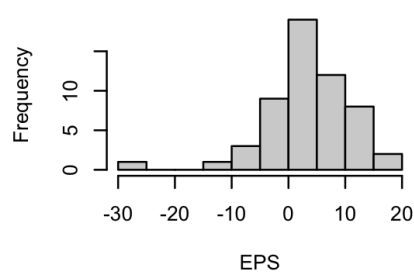


Fig. 3 Hist of BVPS

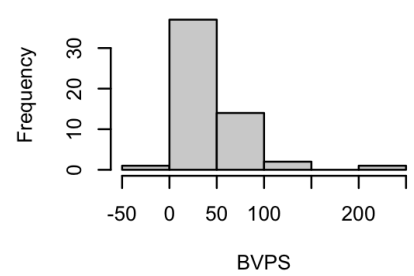


Fig. 4 Hist of CR

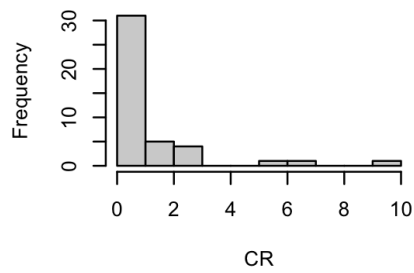
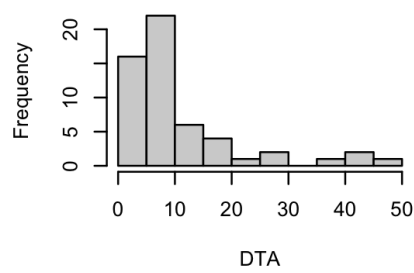
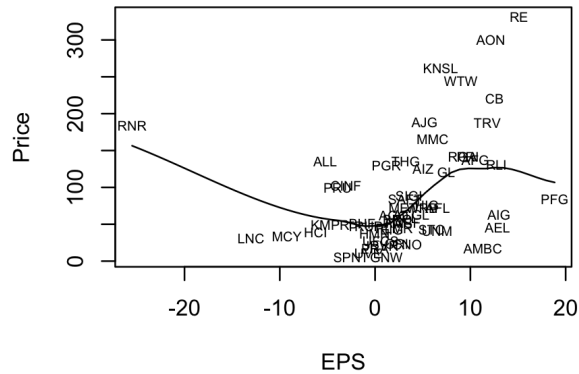
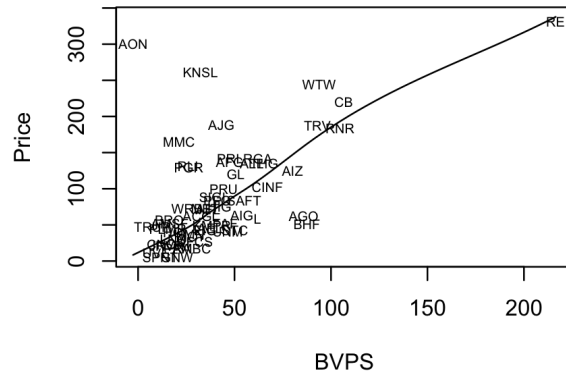
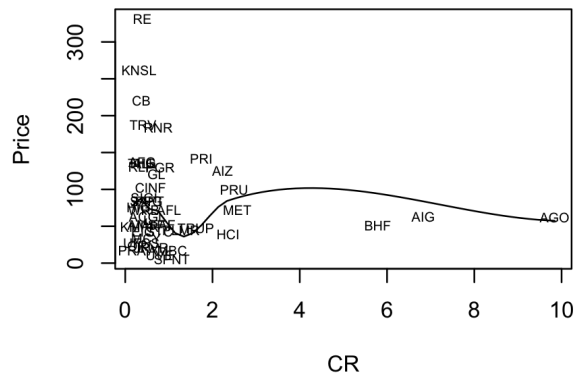
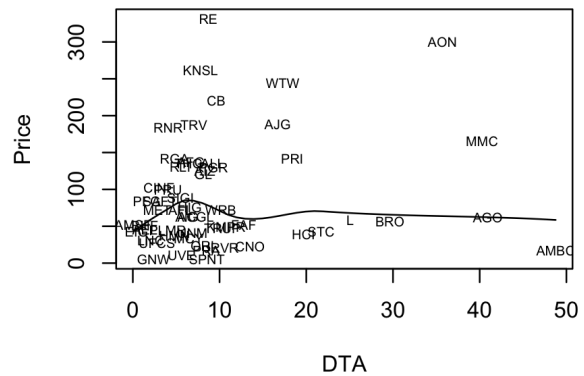


Fig. 5 Hist of DTA



Figures 1 through 5 display the histograms of the frequency of each variable: Price, EPS, BVPS, CR, and DTA. All of the figures, except Figure 2 are skewed to the right and display the presence of potential outliers.

Fig. 6 Scatterplot of Price v EPS**Fig. 7 Scatterplot of Price v BVPS****Fig. 8 Scatterplot of Price v CR****Fig. 9 Scatterplot of Price v DTA**

Figures 6 through 9 display the scatterplots of the relationship of each variable (EPS, BVPS, CR, and DTA) with price. All of the figures, except Figure 2 are relatively weak. All of the figures are positive and indicate that the relationship between price and each variable may be nonlinear.

Table 1*Descriptive Statistics*

Name	Obs	Max	Min	Mean	Median	Mode	Var	Std	Skew	Kurt
Price	55	331.27	5.29	92.75	63.24	62.78	5709.07	75.56	1.28	4.28
EPS	55	18.85	-25.50	3.37	3.21	3.80	54.02	7.35	-1.02	6.13
BVPS	55	216.44	-2.58	44.45	37.37	32.62	1210.80	34.80	2.38	12.09

CR	55	9.84	0.16	1.27	0.54	0.51	3.57	1.89	4.46	13.03
DTA	55	48.76	0.03	10.66	7.26	7.26	113.48	10.65	1.92	6.43

Table 1 demonstrates the descriptive statistics.

Table 2

Correlation Matrix

	Price	EPS	BVPS	CR	DTA
Price	1.000	0.265	0.731	-0.148	-0.147
EPS	0.265	1.000	0.164	-0.007	0.124
BVPS	0.731	0.164	1.000	0.160	0.038
CR	-0.148	-0.007	0.160	1.000	0.331
DTA	-0.147	0.124	0.038	0.331	1.000

Table 2 demonstrates the correlation between the different variables. The variable BVPS has a strong, positive correlation with price, while the other variables have a weak correlation with price. There is no multicollinearity between variables.

Table 3

Regression Results

	Estimate	Std. Error	T Value	Pr(> t)
(Intercept)	36.1579	12.8035	2.824	0.00751
EPS	1.4206	0.9159	1.551	0.12916
BVPS	1.4073	0.1894	7.432	6.45e-09
CR	-8.2835	3.8303	-2.163	0.03693
DTA	-0.8578	0.7551	-1.136	0.26308

*Residual standard error: 43.56 on 38 degrees of freedom

*Multiple R-squared: 0.6377, Adjusted R-squared: 0.5996

*F-statistic: 16.72 on 4 and 38 DF, p-value: 5.489e-08

Eqn 4 (Regression line) $\text{Price} = 36.15 + 1.42 \text{ EPS} + 1.40 \text{ BVPS} - 8.28 \text{ CR} - 0.85 \text{ DTA}$

Table 3 demonstrates the regression coefficients and has Eqn 4 or the regression line. EPS, BVPS, and CR all have significant coefficients. Additionally, 63.77% of the total variation of price is explained by the variation of the independent variables. The model is also significant.

Fig. 10 Scatterplot of EPS Using GAM

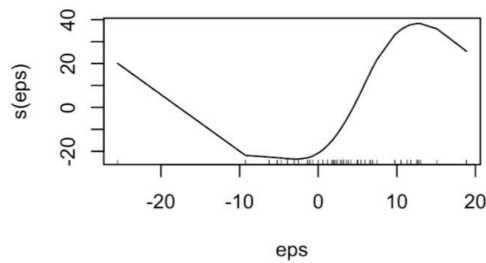


Fig. 11 Scatterplot of CR Using GAM

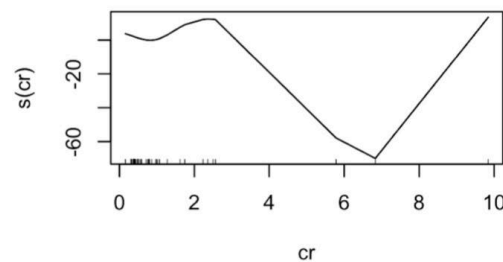


Fig. 12 Scatterplot of BVPS Using GAM

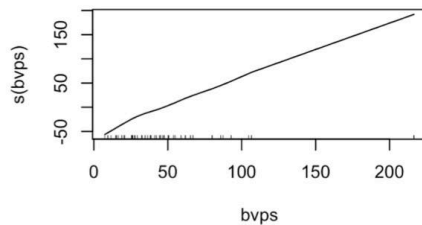
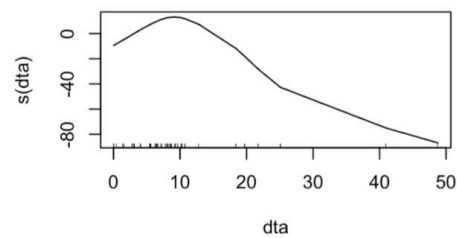


Fig. 13 Scatterplot of DTA Using GAM



Figures 10 through 13 demonstrate the scatterplots of the variables using multiple nonlinear, nonparametric regression, which is from the package generalized additive model (gam) in R. Figures 10 through 12 display a nonlinear positive relationship with price. Figure 13 shows a nonlinear negative relationship with price.

VI. Conclusion

In conclusion, the study was successful. The model was significant (f-statistic= 16.72) and there was an explanatory power of 63.77%. The variables EPS, BVPS, and CR were predictive of stock price. Future research might expand by analyzing more variables.

VII. Bibliography

Data obtained from: Dr. Russon's course materials; S&P1500

VIII. Appendix I

	tkr	price	eps	bvps	cr	dta
351	ACGL	62.78	3.8034	32.61834	0.5068626	7.26083824
768	AEL	45.62	12.861	37.36806	NA	1.39868638
866	AFG	137.28	10.5276	47.55645	0.3903897	6.69157327
890	AFL	71.94	6.5882	36.35072	1.0044463	5.5178425
940	AGO	62.26	1.9555	85.81154	9.8417508	40.9428249
1047	AIG	63.24	13.0048	53.82848	6.8301557	6.23658088
1151	AIZ	125.06	5.0491	80.04296	2.2305285	8.30694658
1177	AJG	188.54	5.1896	43.15054	NA	16.6997363
1451	ALL	135.6	-5.2212	58.95437	0.4409759	9.21505114
1601	AMBC	17.44	11.311	27.83846	0.9731654	48.7587277
1900	AMSF	51.97	2.8777	16.571	0.485439	0.02914064
2172	AON	300.14	12.1435	-2.57546	NA	35.7051125
4061	BHF	51.27	-1.3567	87.48051	5.7856203	1.38883388
4742	BRO	56.97	2.3658	16.26624	NA	29.6458515
5321	CB	220.6	12.5446	106.61972	0.3695545	9.61741256
6410	CINF	102.39	-3.0604	67.03374	0.5805669	3.02327959
6924	CNO	22.85	3.3708	12.25085	NA	13.5149068
10167	EIG	43.13	1.7485	34.7634	0.407165	0.42589438
11423	FAF	52.34	2.4511	45.20349	0.8603537	12.8050819
13255	GL	120.55	7.4729	50.6085	0.7131103	8.13346152
13478	GNW	5.29	1.1918	20.1697	NA	2.36799362
14149	HCI	39.59	-6.2434	18.90941	2.3572732	19.6917486
14379	HIG	75.83	5.4446	42.1978	0.3082533	6.56550435
14521	HMN	37.37	-0.0625	26.60355	NA	4.77600679
16907	JRV	20.91	0.5902	14.77882	0.5925403	10.2021334
17399	KMPR	49.2	-4.7191	38.22711	0.329227	10.745761
17484	KNSL	261.52	6.8806	32.28376	0.3145236	7.80985241
17776	L	58.33	4.1598	61.86616	0.5004342	25.0877594
18337	LNC	30.72	-13.0208	18.40793	NA	2.07791473
19436	MCY	34.2	-9.2588	27.48968	0.4851649	6.4458694
19695	MET	72.37	2.9101	34.70678	2.5659109	2.80463419
20015	MMC	165.48	6.0396	21.26173	NA	40.2702218
23261	ORI	24.15	2.2631	20.76635	0.3201814	8.02790917
24210	PPG	83.92	18.8468	41.07474	0.5415777	1.61252359
24315	PGR	129.71	1.1831	26.32433	0.8009422	9.29109732
24619	PLMR	45.16	2.02	15.37327	1.2789523	4.02026034
25088	PRA	17.47	-0.0074	20.45855	0.157225	8.50099684
25244	PRI	141.82	9.7447	46.74938	1.7449368	18.3900987
25301	PRU	99.46	-3.9296	44.40343	2.4983063	4.06077833
25920	RE	331.27	15.1026	216.4359	0.3934514	8.70537519
26103	RGA	142.09	9.2024	62.1661	NA	4.8161568
26346	RLI	131.27	12.7399	25.89284	0.3105736	5.45061211
26436	RNR	184.23	-25.5032	104.65463	0.7567859	4.07738756
27023	SAFT	84.26	3.1651	54.88063	0.5009983	3.1339313
27813	SIGI	88.61	3.5431	38.57485	0.4333761	5.51199714
28615	SPNT	5.9	-2.5139	11.55955	1.0668035	8.54875259
28965	STC	42.73	5.935	50.20753	0.7849975	21.7324798
30024	THG	135.13	3.2133	65.32584	0.3892583	6.42412001
30712	TRUP	47.53	-1.0958	7.44388	1.6197171	10.3416033
30734	TRV	187.49	11.773	92.891	0.4078425	7.04736725
31355	UFCS	27.36	0.5937	29.35732	0.3730626	2.78769535
31624	UNM	41.03	6.5024	46.50968	NA	6.87164173
31920	UVE	10.59	-0.7241	9.47369	0.7872976	5.6174561
33567	WRB	72.57	4.9419	25.5091	0.4292559	10.1454183
33853	WTW	244.58	9.0089	93.8211	NA	17.2841449

Showing 35 to 55 of 55 entries, 6 total columns

*Original data (spInfo and spData) too large to paste, so this is the idf (industry dataframe), which was used for all analysis.

IX. Appendix II

```
# Leslie Alhakim
```

```
# BUA 633 Term Project 1
```

```
# Import packages
```

```
library(YRmisc)
```

```
# Import datasets
```

```
library(readxl)
```

```
spInfo <- read_excel("Downloads/BUA 633/spInfo-1 (1).xlsx")
```

```
View(spInfo)
```

```
spData <- read_excel("Downloads/BUA 633/spData-1 (1).xlsx")
```

```
View(spData)
```

```
# Preliminaries
```

```
names(spData)
```

```
names(spInfo)
```

```
data.class(spData)
```

```
spData<-as.data.frame(spData)
```

```
data.class(spData)
```

```
data.class(spInfo)
```

```
spInfo<-as.data.frame(spInfo)
```

```
data.class(spInfo)
```

```
# Merge dataframes
```

```
spdf<-merge(spData,spInfo,by="tkr")
```

```
dim(spdf)
```

```
names(spdf)
```

```
# Extract "year" variable
```

```
spdf$date<-as.numeric(substr(spdf$date,7,10))
```

```
names(spdf)[2]<-"year"
```

```
names(spdf)
```

```
# Cross section analysis
```

```
max(spdf$year)
```

```
csdf<-spdf[spdf$year==2022,]
```

```
names(csdf);dim(csdf);
```

```
data.frame(table(csdf$industry))
```

```
# Choose an industry: INSURANCE
```

```
idf<-csdf[csdf$industry=="Insurance",
```

```

      c("tkr", "price", "eps", "bvps", "cr", "dta")]
idf; dim(idf)

# Graphical techniques: HISTOGRAMS

histogram <-function(x,myTitle,xxlab,yylab){
  hist(x,main=myTitle,xlab=xxlab,ylab=yylab) }

par(mfrow=c(3,3))

histogram(idf$price,"Fig. 1 Hist of Price", "Price","Frequency")
histogram(idf$eps,"Fig. 2 Hist of EPS", "EPS","Frequency")
histogram(idf$bvps,"Fig. 3 Hist of BVPS", "BVPS","Frequency")
histogram(idf$cr,"Fig. 4 Hist of CR", "CR","Frequency")
histogram(idf$dta,"Fig. 5 Hist of DTA", "DTA","Frequency")

# Graphical techniques: SCATTERPLOTS

plotTkr<-function(x,y,z,xxlab,yylab,myTitle){

  scatter.smooth(x,y,type="n",xlab=xxlab,ylab=yylab,main=myTitle)

  text(x,y,z,cex=.6)  }

par(mfrow=c(2,2))

plotTkr(idf$eps,idf$price,idf$tkr,"EPS","Price","Fig. 6
Scatterplot of Price v EPS")

```

```

plotTkr(idf$bvps,idf$price,idf$tkr,"BVPS","Price","Fig. 7
Scatterplot of Price v BVPS")

plotTkr(idf$cr,idf$price,idf$tkr,"CR","Price","Fig. 8
Scatterplot of Price v CR")

plotTkr(idf$dta,idf$price,idf$tkr,"DTA","Price","Fig. 9
Scatterplot of Price v DTA")

# Analytical methods: DESCRIPTIVE STATISTICS

ds.summ(idf[,c("price","eps","bvps","cr","dta")],2)

# Analytical methods: CORRELAION

round(cor(idf[,c("price","eps","bvps","cr","dta")]),3)

# Analytical methods: REGRESSION RESULTS

fit<-lm(price ~ eps+bvps+cr+dta,na.action=na.omit,data=idf)

summary(fit)

# Post regression validation

par(mfcol=c(2,2))

hist(fit$residuals, main ="Fig. 10 Hist of Residuals")

plotTkr(fit$fitted.values,fit$model$price,idf$tkr, "Predicted
Price","Actual Price","Fig. 11 Act v Pred Price")

# Graphical methods: NONLINEAR, NONPARAMETRIC REGRESSION

```

```
library(gam)

fit<-gam(price~eps+bvps+cr+dta,na.action=na.omit,data=idf)

cor(fit$model$price,fit$fitted.values)^2
```